

1. Project Overview

- **Objective:** Design a robust, scalable, and secure network for a new building with 600 users across 3 floors.
- **Key Components:**
 - **Floors:**
 - **First Floor:** Sales & Marketing (120 users), Human Resource & Logistics (120 users).
 - **Second Floor:** Finance & Accounts (120 users), Administration & Public Relations (120 users).
 - **Third Floor:** ICT (120 users), Server Room (12 devices).
 - **Network Requirements:**
 - Hierarchical design with redundancy.
 - Wireless networking for each department.
 - VLAN segmentation and subnetting.
 - Dynamic IP addressing via DHCP (except server room).
 - Inter-VLAN routing using multilayer switches.
 - OSPF as the routing protocol.
 - SSH for remote access.
 - Port security for Finance & Accounts.
 - PAT with ACLs for internet access.
 - Connectivity to two ISPs for redundancy.

2. Hierarchical Network Design

- **Core Layer:**
 - **Routers:** R1 and R2 for redundancy and ISP connectivity.
- **Distribution Layer:**
 - **Multilayer Switches:** MLS-W1 and MLS-W2 for inter-VLAN routing and redundancy.
- **Access Layer:**
 - **Access Switches:** Connect end-user devices.

3. Redundancy

- **Router Redundancy:** Two routers (R1 and R2) ensure high availability.
- **Switch Redundancy:** Two multilayer switches (MLS-W1 and MLS-W2) provide redundancy.
- **ISP Redundancy:** Two ISPs ensure internet connectivity even if one fails.

4. VLAN Segmentation

- Each department is assigned a unique VLAN:
 - **VLAN 10:** Sales & Marketing.
 - **VLAN 20:** Human Resource & Logistics.
 - **VLAN 30:** Finance & Accounts.
 - **VLAN 40:** Administration & Public Relations.
 - **VLAN 50:** ICT.
 - **VLAN 60:** Server Room.

5. Subnetting

- Base network: **172.16.1.0**.
- Subnets:
 - **First Floor:**
 - Sales & Marketing: **172.16.1.0/25**.
 - Human Resource & Logistics: **172.16.1.128/25**.
 - **Second Floor:**
 - Finance & Accounts: **172.16.2.0/25**.
 - Administration & Public Relations: **172.16.2.128/25**.
 - **Third Floor:**
 - ICT: **172.16.3.0/25**.
 - Server Room: **172.16.3.128/28**.

6. Dynamic IP Addressing

- **DHCP Servers:** Located in the server room.
 - Assign IP addresses dynamically to all devices except server room devices (static IPs).

7. Inter-VLAN Routing

- **Multilayer Switches:** MLS-W1 and MLS-W2 handle inter-VLAN routing.
 - Allow devices in different VLANs to communicate.

8. Routing Protocol

- **OSPF (Open Shortest Path First):**
 - Used for routing between routers and switches.
 - Ensures efficient route advertisement and redundancy.

9. SSH for Remote Access

- **SSH (Secure Shell):**
 - Configured on all routers and multilayer switches.
 - Provides secure remote access for network management.

10. Port Security

- **Finance & Accounts Department:**
 - Only one device per switch port.
 - **Sticky Method:** Learns and stores MAC addresses.
 - **Violation Mode:** Shuts down the port if unauthorized devices connect.

11. PAT with ACLs

- **PAT (Port Address Translation):**
 - Uses the outbound router interface IPv4 address.
 - **ACLs (Access Control Lists):**
 - Control internet access by defining rules for traffic.

12. Wireless Networking

- Each department has its own wireless network.
- **Wireless Access Points:**
 - Provide wireless connectivity to users.

13. Technologies Used

- **Cisco Packet Tracer:** For network design and simulation.
- **Routers:** Cisco routers (R1 and R2).
- **Multilayer Switches:** Cisco MLS-W1 and MLS-W2.
- **Access Switches:** For connecting end-user devices.
- **Wireless Access Points:** For wireless networking.
- **Protocols:**
 - **OSPF:** Routing.
 - **SSH:** Secure remote access.
 - **PAT:** Internet access translation.
 - **DHCP:** Dynamic IP allocation.
 - **VLANs:** Network segmentation.

14. Key Configurations

- **Basic Device Settings:**
 - Hostnames, console password, enable password, banner messages, disable IP domain lookup.
- **VLAN Assignment:**
 - VLANs assigned to each department.
- **Switchport Security:**
 - Configured for Finance & Accounts department.
- **OSPF Configuration:**

- On routers and multilayer switches.
- **Static IP for Server Room:**
 - Devices in the server room have static IPs.
- **DHCP Server Configuration:**
 - DHCP pools for each department.
- **Inter-VLAN Routing:**
 - Configured on multilayer switches.
- **PAT + ACL:**
 - Configured on routers for internet access.

15. Testing and Verification

- **Ping Tests:**
 - Ensure devices can communicate across VLANs and with the internet.
- **SSH Access:**
 - Verify secure remote access to routers and switches.
- **Port Security:**
 - Test by connecting unauthorized devices to Finance & Accounts ports.
- **PAT and ACL:**
 - Verify internet access and traffic control.

16. Conclusion

- The network design ensures **high availability, security, and scalability**.
- It meets current business needs and is **future-proofed** for growth.
- The use of **redundancy, VLANs, OSPF, and SSH** ensures a robust and secure network.

Key Points to Remember

1. **Hierarchical Design:** Core, Distribution, and Access layers.
2. **Redundancy:** Routers, switches, and ISPs.
3. **VLANs:** Each department has its own VLAN.
4. **Subnetting:** Efficient IP allocation using 172.16.1.0.
5. **DHCP:** Dynamic IP assignment for all devices except server room.
6. **Inter-VLAN Routing:** Handled by multilayer switches.
7. **OSPF:** Routing protocol for efficient route advertisement.
8. **SSH:** Secure remote access to devices.
9. **Port Security:** Finance & Accounts department.
10. **PAT + ACL:** Internet access and traffic control.
11. **Wireless Networking:** Each department has its own wireless network.

Common Viva Questions

1. **What is the purpose of VLANs in this network?**
 - VLANs segment the network by department, improving security and reducing broadcast traffic.
2. **Why is OSPF used instead of other routing protocols?**
 - OSPF is efficient, scalable, and supports fast convergence, making it ideal for large networks.
3. **How does PAT work in this network?**
 - PAT allows multiple devices to share a single public IP address for internet access, conserving IP addresses.
4. **What is the role of multilayer switches?**
 - They handle both switching and routing, enabling inter-VLAN communication.
5. **Why is port security important for the Finance & Accounts department?**
 - It ensures only authorized devices can connect, enhancing security.
6. **How does DHCP simplify IP address management?**
 - DHCP automatically assigns IP addresses to devices, reducing manual configuration errors.
7. **What is the significance of having two ISPs?**
 - It provides redundancy, ensuring internet connectivity even if one ISP fails.
8. **How do you test the network after configuration?**
 - Use ping tests, SSH access, and verify port security and PAT functionality.

Final Tips

- **Practice Configurations:** Be familiar with basic commands for VLANs, OSPF, SSH, and PAT.
- **Understand Subnetting:** Be able to explain how subnetting works and why it's used.
- **Explain Redundancy:** Be clear on how redundancy improves network reliability.

1. What is the purpose of using multilayer switches in this network?

- Multilayer switches handle both switching (Layer 2) and routing (Layer 3), enabling inter-VLAN communication and reducing the need for separate routers.

2. Why is OSPF preferred over static routing in this network?

- OSPF dynamically adjusts to network changes, provides faster convergence, and reduces manual configuration errors compared to static routing.

3. What is the role of the DHCP server in this network?

- The DHCP server automatically assigns IP addresses to devices, ensuring efficient IP management and reducing manual configuration.

4. How does VLAN segmentation improve network performance?

- VLANs reduce broadcast traffic by isolating traffic within departments, improving network efficiency and security.

5. What is the purpose of the "sticky" method in port security?

- The sticky method dynamically learns and stores the MAC address of the first device connected to a port, ensuring only that device can connect in the future.

6. How does PAT help in conserving IP addresses?

- PAT allows multiple devices to share a single public IP address for internet access, reducing the need for multiple public IPs.

7. What is the significance of having two ISPs in this network?

- Two ISPs provide redundancy, ensuring internet connectivity remains available even if one ISP fails.

8. How do you ensure secure remote access to network devices?

- SSH (Secure Shell) is configured on routers and switches to provide encrypted remote access, preventing unauthorized access.

9. What is the difference between a router and a switch?

- **Router:** Operates at Layer 3 (Network Layer), routes data between different networks, and uses IP addresses.
- **Switch:** Operates at Layer 2 (Data Link Layer), forwards data within the same network, and uses MAC addresses.

10. What is the purpose of the server room VLAN?

- The server room VLAN isolates critical server devices, providing enhanced security and management for server resources.

11. How do you verify that OSPF is working correctly?

- Use commands like `show ip ospf neighbor` to check OSPF neighbor relationships and `show ip route` to verify OSPF routes.

12. What is the purpose of the "ip helper-address" command?

- The ip helper-address command forwards DHCP requests from clients to the DHCP server, enabling dynamic IP assignment across VLANs.

13. How do you handle a port security violation?

- The port is shut down if an unauthorized device connects, preventing unauthorized access.

14. What is the difference between a VLAN and a subnet?

- **VLAN:** A logical grouping of devices within a network, regardless of physical location.
- **Subnet:** A division of an IP network, defined by a range of IP addresses.

15. Why is inter-VLAN routing necessary?

- Inter-VLAN routing allows devices in different VLANs to communicate with each other, which is essential for departments to share resources.

16. What is the purpose of the "banner motd" command?

- The banner motd command displays a message when someone logs into the device, often used for security warnings.

17. How do you ensure that the network is future-proofed?

- By using scalable technologies like VLANs, OSPF, and multilayer switches, the network can accommodate future growth and changes.

18. What is the difference between a public IP and a private IP?

- **Public IP:** Used for devices accessible over the internet.
- **Private IP:** Used for devices within a local network, not accessible directly from the internet.

19. How do you configure a VLAN on a switch?

- Use the following commands:

```
Copy
Switch(config)# vlan 10
Switch(config-vlan)# name Sales
Switch(config-vlan)# exit
Switch(config)# interface fa0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
```

20. What is the purpose of the "no ip domain-lookup" command?

- The no ip domain-lookup command disables DNS lookup, preventing the device from trying to resolve unknown commands as hostnames.

What is Subnetting?

- **Subnetting** is the process of dividing a larger IP network into smaller, more manageable subnetworks (subnets).
- **Purpose:**
 - Efficient use of IP addresses.
 - Reduce network congestion by limiting broadcast traffic.
 - Improve security by isolating network segments.
- **Example:**
 - A network with the base IP 172.16.1.0/24 can be divided into smaller subnets like 172.16.1.0/25 and 172.16.1.128/25.

Difference Between Router and Switch

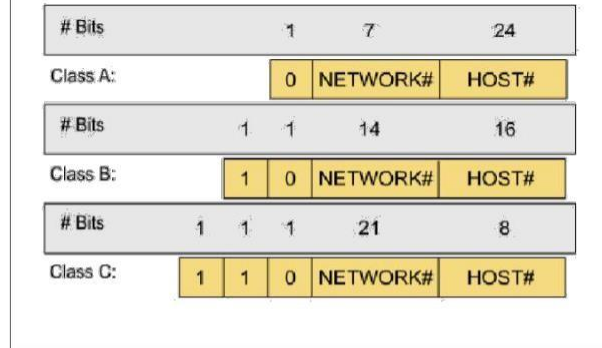
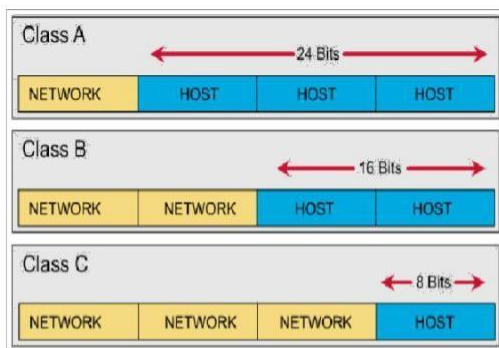
Feature	Router	Switch
Layer	Operates at Layer 3 (Network Layer) .	Operates at Layer 2 (Data Link Layer) .
Function	Routes data between different networks.	Forwards data within the same network.
Addressing	Uses IP addresses to route packets.	Uses MAC addresses to forward frames.
Broadcast Domain	Breaks broadcast domains (each port is a separate broadcast domain).	All ports are in the same broadcast domain (unless VLANs are used).
Use Case	Connects different networks (e.g., LAN to WAN).	Connects devices within the same network (e.g., PCs, printers).
Example	Connects your home network to the internet.	Connects devices in an office LAN.

Key Differences in a Nutshell

- **Router:** Connects different networks, uses IP addresses, and operates at Layer 3.
- **Switch:** Connects devices within the same network, uses MAC addresses, and operates at Layer 2.

Final Tips for Viva

1. **Understand the Basics:** Be clear on concepts like VLANs, subnetting, routing, and switching.
2. **Practice Configurations:** Be familiar with basic commands for VLANs, OSPF, SSH, and PAT.
3. **Explain Redundancy:** Be able to explain how redundancy improves network reliability.



Class	Starts with	Binary range	Decimal Value range	Maximum subnets	Maximum hosts	Routing mask
A	0	00000000-01111111	0-127*	127	16,777,214	255.0.0.0
B	10	10000000-10111111	128-191	16,384	65,534	255.255.0.0
C	110	11000000-11011111	192-223	2,097,152	254	255.255.255.0
D	1110	11100000-11101111	224-239			
E	1111	11110000-11111111	240-255			

1. IP Addressing

- **Definition:** An IP address uniquely identifies a device on an IP network.
- **Format:** 32-bit address divided into 4 octets (e.g., 172.16.81.100).
- **Binary to Decimal:** Each octet ranges from 0 to 255 (e.g., 205.112.45.60 = 11001101.01110000.00101101.00111100).
- **Parts of an IP Address:**
 - **Network Portion:** Identifies the network.
 - **Host Portion:** Identifies the device within the network.

2. Classful Addressing

- **Classes:** IPv4 addresses are divided into 5 classes:
 - **Class A:** 0.0.0.0 to 127.255.255.255 (Default mask: 255.0.0.0 or /8).
 - **Class B:** 128.0.0.0 to 191.255.255.255 (Default mask: 255.255.0.0 or /16).
 - **Class C:** 192.0.0.0 to 223.255.255.255 (Default mask: 255.255.255.0 or /24).
 - **Class D:** 224.0.0.0 to 239.255.255.255 (Used for multicast).
 - **Class E:** 240.0.0.0 to 255.255.255.255 (Reserved for experimental use).

3. Subnetting

- **Definition:** Dividing a large network into smaller subnets.
- **Purpose:**
 - Efficient IP address usage.
 - Reduce network congestion.
 - Improve security by isolating segments.
- **Subnet Mask:** 32-bit mask that separates the network and host portions.
 - **Network Bits:** Set to 1.
 - **Host Bits:** Set to 0.
 - Example: IP = 192.168.1.10, Subnet Mask = 255.255.255.0 → Network = 192.168.1, Host = 10.

4. CIDR (Classless Inter-Domain Routing)

- **Definition:** Allows flexible allocation of IP addresses by using variable-length subnet masks (VLSM).
- **Slash Notation:** Represents the number of network bits (e.g., /24 for Class C).
- **Example:**
 - Class A: /8 (255.0.0.0).
 - Class B: /16 (255.255.0.0).
 - Class C: /24 (255.255.255.0).

5. Subnetting Class C Addresses

- **Default Mask:** 255.255.255.0 (/24).
- **Host Bits:** 8 bits available for hosts.
- **Example:**

- IP: 192.168.1.0/26 (borrowed 2 bits for subnets).
- Subnets: 4 (2^2).
- Hosts per subnet: 64 ($2^6 - 2$).

6. Subnetting Class B Addresses

- **Default Mask:** 255.255.0.0 (/16).
- **Host Bits:** 16 bits available for hosts.
- **Example:**
 - IP: 172.16.0.0/20 (borrowed 4 bits for subnets).
 - Subnets: 16 (2^4).
 - Hosts per subnet: 4094 ($2^{12} - 2$).

7. Subnetting Class A Addresses

- **Default Mask:** 255.0.0.0 (/8).
- **Host Bits:** 24 bits available for hosts.
- **Example:**
 - IP: 10.0.0.0/12 (borrowed 4 bits for subnets).
 - Subnets: 16 (2^4).
 - Hosts per subnet: 1,048,574 ($2^{20} - 2$).

8. Key Formulas

- **Number of Subnets:** 2^x (x = borrowed host bits).
- **Total Hosts per Subnet:** 2^y (y = remaining host bits).
- **Valid Hosts per Subnet:** $2^y - 2$ (subtract network and broadcast addresses).
- **Block Size:** $256 - \text{subnet mask}$ (e.g., $256 - 192 = 64$).

9. Example Calculations

- IP: 192.168.1.0/26
 - **Subnet Mask:** 255.255.255.192.
 - **Subnets:** 4 (2^2).
 - **Hosts per Subnet:** 64 ($2^6 - 2$).
 - **Valid Host Range:** 192.168.1.1 to 192.168.1.62.
 - **Broadcast Address:** 192.168.1.63.

10. Key Terms

- **Network Address:** First address of a subnet (e.g., 192.168.1.0).
- **Broadcast Address:** Last address of a subnet (e.g., 192.168.1.63).
- **Valid Hosts:** Addresses between network and broadcast addresses.

11. Summary

- **IP Addressing:** Identifies devices on a network.
- **Subnetting:** Breaks large networks into smaller subnets.
- **CIDR:** Allows flexible IP allocation using VLSM.
- **Classful Addressing:** Divides IP space into classes (A, B, C, D, E).
- **Key Formulas:** Subnets, hosts, block size, and valid hosts.

Quick Tips for Viva

1. **Understand Binary to Decimal Conversion:** Essential for subnetting.
2. **Memorize Key Formulas:** Subnets, hosts, and block size.
3. **Practice Examples:** Work through subnetting problems.
4. **Explain Concepts Clearly:** Be ready to explain subnetting and CIDR.

Here’s a concise explanation of the terms and the process of calculating subnet masks:

Key Concepts

- 1. **Network Address:**
 - The first IP address in a subnet.
 - Used to identify the subnet itself.
 - **Example:** For 172.16.1.0/25, the network address is 172.16.1.0.
- 2. **Host Range:**
 - The range of usable IP addresses in a subnet that can be assigned to devices (hosts).
 - Excludes the **Network Address** and the **Broadcast Address**.
 - **Example:** For 172.16.1.0/25, the host range is 172.16.1.1 to 172.16.1.126.
- 3. **Broadcast Address:**
 - The last IP address in a subnet.
 - Used to send messages to all devices within the subnet.
 - **Example:** For 172.16.1.0/25, the broadcast address is 172.16.1.127.

How Subnet Masks Work

A **Subnet Mask** divides an IP address into two parts:

- 1. **Network Part:** Identifies the network.
- 2. **Host Part:** Identifies devices (hosts) within the network.

Example: 255.255.255.128 (/25)

- **/25** means the first 25 bits are for the network, and the remaining 7 bits are for hosts.
- Total IPs in a subnet = $2^7=1282^7 = 128$ (since 7 host bits remain).
- Usable IPs = $128-2=126$ (subtract 2 for Network and Broadcast addresses).

Subnet Mask Calculation

The subnet mask determines how many IPs are available in a subnet. Common subnet masks:

CIDR	Subnet Mask	Total IPs	Usable IPs	Purpose
/24	255.255.255.0	256	254	Standard subnet for large networks.
/25	255.255.255.128	128	126	Divides /24 into two equal subnets.
/28	255.255.255.240	16	14	Small subnets for specific needs.
/30	255.255.255.252	4	2	Point-to-point links.

Steps to Calculate a Subnet

- 1. **Decide the number of hosts needed:**
 - Each department or connection needs enough IPs for devices.
- 2. **Choose the closest subnet mask:**
 - Match the number of required hosts to a subnet that provides enough usable IPs.
- 3. **Assign subnets sequentially:**
 - Start from the base network and increment by the subnet size.

Worked Example

Given:

- Base network: 172.16.1.0
- Required hosts for **Sales & Marketing**: Up to 126.

Step 1: Choose subnet mask

- Closest subnet: /25 (128 total IPs, 126 usable).

Step 2: Calculate details for 172.16.1.0/25

- **Network Address:** 172.16.1.0
- **Host Range:** 172.16.1.1 to 172.16.1.126
- **Broadcast Address:** 172.16.1.127

Step 3: Move to the next subnet for HR

- Start after broadcast address: **172.16.1.128/25**.

Short Notes on VLAN Configuration and Verification

1. Displaying VLANs on S1

- **Command:** show vlan brief
 - Displays all VLANs configured on the switch.

2. Creating and Naming VLANs on S1

- **Steps:**
 1. Enter global configuration mode: S1# configure terminal
 2. Create VLAN 10: S1(config)# vlan 10
 3. Name VLAN 10: S1(config-vlan)# name Faculty/Staff
- **Repeat for Other VLANs:**
 - VLAN 20: name Students
 - VLAN 30: name Guest(Default)
 - VLAN 99: name Management&Native
 - VLAN 150: name VOICE

3. Creating VLANs on S2 and S3

- **Steps:**
 - Repeat the same VLAN creation process on S2 and S3.

4. Assigning VLANs to Ports on S2

- **Steps:**
 1. Enter interface configuration mode: S2(config)# interface f0/11
 2. Set port mode to access: S2(config-if)# switchport mode access
 3. Assign VLAN 10: S2(config-if)# switchport access vlan 10
- **Repeat for Other Ports:**
 - VLAN 20: interface f0/18, switchport access vlan 20
 - VLAN 30: interface f0/6, switchport access vlan 30

5. Assigning VLANs to Ports on S3

- **Steps:**
 - Repeat the same VLAN assignment process on S3.
 - **VOICE VLAN:**
 1. Enter interface configuration mode: S3(config)# interface f0/11
 2. Enable QoS trust: S3(config-if)# mls qos trust cos
 3. Assign VOICE VLAN: S3(config-if)# switchport voice vlan 150

6. Verifying Connectivity

- **Command:** ping between PC1 (172.17.10.21) and PC4 (172.17.20.25).
- **Expected Result:** Pings should fail because PC1 and PC4 are in different VLANs (VLAN 10 and VLAN 20), and inter-VLAN routing is not configured.

7. Explanation of Ping Failure

- **Reason:** VLANs segment the network, and devices in different VLANs cannot communicate without a Layer 3 device (router or Layer 3 switch) to route traffic between VLANs.
- **Solution:** Configure inter-VLAN routing to allow communication between VLANs.

Key Commands Summary

1. **Display VLANs:** show vlan brief
2. **Create VLANs:**
 - vlan <ID>
 - name <VLAN_NAME>
3. **Assign VLANs to Ports:**
 - interface <port>
 - switchport mode access

- switchport access vlan <ID>
- 4. **Assign VOICE VLAN:**
 - mls qos trust cos
 - switchport voice vlan <ID>
- 5. **Verify Connectivity:** ping <IP>

Quick Tips for Viva

1. **Understand VLAN Segmentation:** VLANs isolate traffic, and devices in different VLANs cannot communicate without routing.
2. **Practice Commands:** Be familiar with VLAN creation, assignment, and verification commands.
3. **Explain Inter-VLAN Routing:** Be ready to explain why pings fail between VLANs and how to fix it.

What is an Interface?

1. Definition

- An **interface** is a physical or logical connection point on a network device (such as a router, switch, or computer) that allows the device to communicate with other devices in a network.
- Interfaces are used to send and receive data packets between devices.

2. Types of Interfaces

1. **Physical Interfaces:**
 - **Ethernet Ports:** Used for wired connections (e.g., FastEthernet, GigabitEthernet).
 - Example: FastEthernet 0/1, GigabitEthernet 0/0.
 - **Serial Ports:** Used for WAN (Wide Area Network) connections.
 - Example: Serial 0/0/0.
 - **Wireless Interfaces:** Used for Wi-Fi connections.
 - Example: WLAN0.
2. **Logical Interfaces:**
 - **VLAN Interfaces:** Used for inter-VLAN routing.
 - Example: VLAN 10.
 - **Loopback Interfaces:** Virtual interfaces used for testing and management.
 - Example: Loopback 0.
 - **Tunnel Interfaces:** Used for VPNs (Virtual Private Networks).
 - Example: Tunnel 0.

3. Key Functions of an Interface

- **Connectivity:** Provides a connection point for devices to communicate.
- **Data Transmission:** Sends and receives data packets.
- **Configuration:** Allows network administrators to configure settings like IP addresses, VLANs, speed, and duplex.
- **Security:** Can be configured with access control lists (ACLs) to filter traffic.

4. Interface Configuration

- **Access Mode:** Used for connecting end devices (e.g., PCs, printers). Carries traffic for a single VLAN.
 - Example: switchport mode access.
- **Trunk Mode:** Used for connecting switches. Carries traffic for multiple VLANs.
 - Example: switchport mode trunk.
- **IP Address Assignment:** Assigns an IP address to the interface for routing purposes.
 - Example: ip address 192.168.1.1 255.255.255.0.
- **VLAN Assignment:** Assigns the interface to a specific VLAN.
 - Example: switchport access vlan 10.

5. Example: Configuring an Interface

1. **Enter Global Configuration Mode:**
 - Switch# configure terminal
2. **Enter Interface Configuration Mode:**
 - Switch(config)# interface FastEthernet 0/1

3. **Set Port Mode:**
 - Switch(config-if)# switchport mode access
4. **Assign VLAN:**
 - Switch(config-if)# switchport access vlan 10
5. **Enable the Interface:**
 - Switch(config-if)# no shutdown

6. Common Interface Commands

- **Show Interface Status:**
 - show interface status: Displays the status of all interfaces.
- **Show IP Interface Brief:**
 - show ip interface brief: Displays a summary of IP addresses assigned to interfaces.
- **Enable/Disable Interface:**
 - no shutdown: Enables the interface.
 - shutdown: Disables the interface.

7. Key Points to Remember

- **Physical vs. Logical:** Physical interfaces are hardware-based, while logical interfaces are software-based.
- **Access vs. Trunk:** Access ports carry traffic for a single VLAN, while trunk ports carry traffic for multiple VLANs.
- **IP Addressing:** Interfaces can be assigned IP addresses for routing purposes.
- **VLAN Assignment:** Interfaces can be assigned to VLANs for network segmentation.

8. Quick Tips for Viva

1. **Understand Interface Types:** Know the difference between physical and logical interfaces.
2. **Practice Configuration:** Be familiar with commands like interface, switchport mode access, and ip address.
3. **Explain VLANs:** Be ready to explain how VLANs are assigned to interfaces.

What is a Switchport?

1. Definition

- A **switchport** is a physical interface on a network switch that connects to other devices, such as computers, printers, or other switches.
- It is used to forward data packets between devices within a local area network (LAN).

2. Types of Switchports

1. Access Port:

- **Purpose:** Connects end devices (e.g., PCs, printers) to the switch.
- **Traffic:** Carries traffic for a single VLAN.
- **Configuration:**
 - switchport mode access: Sets the port as an access port.
 - switchport access vlan <VLAN_ID>: Assigns the port to a specific VLAN.
- **Example:**

Copy

```
Switch(config)# interface FastEthernet 0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
```

2. Trunk Port:

- **Purpose:** Connects switches to other switches or routers.
- **Traffic:** Carries traffic for multiple VLANs.
- **Configuration:**
 - switchport mode trunk: Sets the port as a trunk port.
 - switchport trunk allowed vlan <VLAN_LIST>: Specifies which VLANs are allowed on the trunk.
- **Example:**

Copy

```
Switch(config)# interface GigabitEthernet 0/1
```

```
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 10,20,30
```

3. Voice Port:

- **Purpose:** Connects VoIP phones and prioritizes voice traffic.
- **Traffic:** Carries both data and voice traffic (separate VLANs).
- **Configuration:**
 - switchport voice vlan <VLAN_ID>: Assigns a voice VLAN to the port.
 - mls qos trust cos: Enables QoS (Quality of Service) for voice traffic.
- **Example:**

```
Copy
Switch(config)# interface FastEthernet 0/11
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# switchport voice vlan 150
Switch(config-if)# mls qos trust cos
```

3. Key Functions of a Switchport

- **Connectivity:** Provides a connection point for devices to communicate.
- **VLAN Assignment:** Assigns devices to specific VLANs for network segmentation.
- **Traffic Forwarding:** Forwards data packets between devices within the same VLAN or across VLANs (via trunk ports).
- **Security:** Can be configured with port security to restrict access to authorized devices.

4. Common Switchport Commands

- **Show Switchport Status:**
 - show interface status: Displays the status of all switchports.
 - show vlan brief: Displays VLAN assignments for all ports.
- **Enable/Disable Port:**
 - no shutdown: Enables the port.
 - shutdown: Disables the port.
- **Port Security:**
 - switchport port-security: Enables port security.
 - switchport port-security maximum <number>: Sets the maximum number of MAC addresses allowed on the port.
 - switchport port-security violation shutdown: Shuts down the port if a violation occurs.

5. Key Points to Remember

- **Access Ports:** Used for connecting end devices and carry traffic for a single VLAN.
- **Trunk Ports:** Used for connecting switches and carry traffic for multiple VLANs.
- **Voice Ports:** Used for VoIP phones and prioritize voice traffic.
- **VLAN Assignment:** Ensures devices are grouped logically, improving network efficiency and security.

6. Quick Tips for Viva

1. **Understand Port Types:** Know the difference between access, trunk, and voice ports.
2. **Practice Configuration:** Be familiar with commands like switchport mode access, switchport mode trunk, and switchport voice vlan.
3. **Explain VLANs:** Be ready to explain how VLANs are assigned to switchports

Short Notes on VLAN Trunking Protocol (VTP)

1. What is VTP?

- **Definition:** VTP is a Cisco proprietary protocol used to synchronize VLAN information across switches in the same VTP domain.
- **Purpose:** Simplifies VLAN management by propagating VLAN additions, deletions, and changes across the network.

2. VTP Requirements

- **Same VTP Version:** All switches must use the same VTP version.
- **Same Domain Name:** All switches must be part of the same VTP domain.
- **Server Mode:** At least one switch must be in VTP server mode.
- **Matching Authentication:** If a password is set, it must match on all switches.

3. VTP Modes

1. **VTP Server Mode:**
 - **Function:** Can create, modify, and delete VLANs.
 - **Propagation:** Sends VLAN changes to other switches in the domain.
 - **Command:** vtp mode server.
2. **VTP Client Mode:**
 - **Function:** Cannot create, modify, or delete VLANs.
 - **Propagation:** Receives and forwards VLAN updates.
 - **Command:** vtp mode client.
3. **VTP Transparent Mode:**
 - **Function:** Does not participate in VTP but forwards VTP advertisements.
 - **Propagation:** Can create and delete VLANs locally, but changes are not propagated.
 - **Command:** vtp mode transparent.

4. Configuring VTP

1. **Set VTP Mode:**
 - Server: S1(config)# vtp mode server
 - Client: S2(config)# vtp mode client
 - Transparent: S3(config)# vtp mode transparent
2. **Set VTP Domain Name:**
 - S1(config)# vtp domain Lab4
3. **Set VTP Password:**
 - S1(config)# vtp password cisco

5. Configuring Trunking and Native VLAN

1. **Set Trunk Mode:**
 - S1(config)# interface range fa0/1-5
 - S1(config-if-range)# switchport mode trunk
2. **Set Native VLAN:**
 - S1(config-if-range)# switchport trunk native vlan 99
3. **Enable Interface:**
 - S1(config-if-range)# no shutdown

6. Configuring VLANs on VTP Server

1. **Create VLANs:**
 - S1(config)# vlan 99
 - S1(config-vlan)# name management
 - S1(config-vlan)# exit
 - Repeat for VLANs 10, 20, and 30.
2. **Verify VLANs:**
 - S1# show vlan brief

7. Verifying VTP and VLAN Distribution

1. **Check VTP Status:**
 - S1# show vtp status
2. **Verify VLANs on Client Switches:**
 - S2# show vlan brief
 - S3# show vlan brief

8. Key Points to Remember

- **VTP Server:** Manages VLANs and propagates changes.
- **VTP Client:** Receives VLAN updates but cannot make changes.
- **VTP Transparent:** Does not participate in VTP but forwards updates.
- **Trunking:** Allows multiple VLANs to traverse a single link.
- **Native VLAN:** VLAN used for untagged traffic on a trunk port.

Short Notes on Switch Configuration and Verification

1. Enter Privileged EXEC Mode

- **Command:** enable
- **Prompt Change:** Switch> → Switch#

2. Examine Current Switch Configuration

- **Command:** show running-config
- **Questions:**
 - **Fast Ethernet Interfaces:** Check the number of FastEthernet interfaces.
 - **Gigabit Ethernet Interfaces:** Check the number of GigabitEthernet interfaces.

3. Assign a Name to the Switch

- **Commands:**
 - configure terminal
 - hostname S1
 - exit
- **Result:** Prompt changes to S1#.

4. Secure Access to the Console Line

- **Commands:**
 - configure terminal
 - line console 0
 - password letmein
 - login
 - exit
- **Purpose:** Protects console access with a password.

5. Verify Console Access Security

- **Steps:**
 - Exit privileged EXEC mode: exit
 - Press Enter and enter the password when prompted.
- **Result:** Access is secured with the password letmein.

6. Secure Privileged Mode Access

- **Commands:**
 - enable
 - configure terminal
 - enable password c1\$c0
 - exit
- **Purpose:** Protects privileged EXEC mode with a password.

7. Verify Privileged Mode Access Security

- **Steps:**
 - Exit privileged EXEC mode: exit
 - Press Enter and enter the password when prompted.
- **Result:** Access is secured with the password c1\$c0.

8. Configure an Encrypted Password for Privileged Mode

- **Commands:**
 - config t
 - enable secret itsasecret
 - exit
- **Purpose:** Replaces the plaintext enable password with an encrypted enable secret.

9. Configure a Message of the Day (MOTD) Banner

- **Commands:**
 - config t
 - banner motd "This is a secure system. Authorized Access Only!"
 - exit
- **Purpose:** Displays a message when someone logs into the switch.

10. Save Configuration to NVRAM

- **Command:** copy running-config startup-config
- **Purpose:** Saves the current configuration to NVRAM for persistence after reboot.

11. Examine Flash Memory

- **Commands:**
 - show flash
 - dir flash:
- **Purpose:** Displays the contents of the switch's flash memory.

12. Configure an IP Address on PC-A

- **Steps:**
 - Assign IP address: 192.168.1.10/24
 - Assign IPv6 address: 2001:db8:acad:3/64
 - Assign Link-Local address: fe80::3

13. Verify and Test Network Connectivity

- **Commands:**
 - ping <IP Address>: Tests connectivity to another device.
 - show ip interface brief: Verifies IP configuration on the switch.
- **Purpose:** Ensures that the network is functioning correctly.

Key Points to Remember

- **Privileged EXEC Mode:** Accessible via enable command.
- **Configuration Mode:** Entered via configure terminal.
- **Security:**
 - Console access: line console 0, password, login.
 - Privileged mode: enable password, enable secret.
- **MOTD Banner:** banner motd.
- **Saving Configuration:** copy running-config startup-config.
- **Verification:** Use show commands to verify configurations.

Quick Tips for Viva

1. **Understand Modes:** Know the difference between user EXEC, privileged EXEC, and configuration modes.
2. **Practice Commands:** Be familiar with enable, configure terminal, hostname, password, login, enable secret, banner motd, and copy running-config startup-config.
3. **Explain Security:** Be ready to explain how to secure console and privileged mode access.

Short Notes on Inter-VLAN Routing

1. Introduction to Inter-VLAN Routing

- **Purpose:** Enables communication between devices in different VLANs.
- **Problem:** By default, devices in different VLANs cannot communicate because VLANs segment broadcast domains.
- **Solution:** Use a router to forward traffic between VLANs.

2. Methods of Inter-VLAN Routing

1. Traditional Inter-VLAN Routing:

- **Description:** Uses multiple router interfaces, one for each VLAN.
- **Configuration:**
 - Router interfaces are configured as default gateways for each VLAN.
 - Switch ports connected to the router are configured as access ports in their respective VLANs.
- **Example:**
 - VLAN 20: Fa0/0 on router configured as 192.168.20.1/24.
 - VLAN 30: Fa0/1 on router configured as 192.168.30.1/24.

2. Router-on-a-Stick:

- **Description:** Uses a single router interface with subinterfaces for each VLAN.
- **Configuration:**
 - Router interface is configured as a trunk port.
 - Subinterfaces are created for each VLAN.
- **Example:**
 - VLAN 20: Subinterface Fa0/0.20 configured as 192.168.20.1/24.
 - VLAN 30: Subinterface Fa0/0.30 configured as 192.168.30.1/24.

3. Traditional Inter-VLAN Routing Configuration

- **Switch Configuration:**
 - Assign switch ports to VLANs:
 - interface Fa0/1
 - switchport mode access
 - switchport access vlan 20
 - interface Fa0/2
 - switchport mode access
 - switchport access vlan 30
- **Router Configuration:**
 - Configure router interfaces as default gateways:
 - interface Fa0/0
 - ip address 192.168.20.1 255.255.255.0
 - no shutdown
 - interface Fa0/1
 - ip address 192.168.30.1 255.255.255.0
 - no shutdown

4. Router-on-a-Stick Configuration

- **Switch Configuration:**
 - Configure trunk port:
 - interface Fa0/1
 - switchport mode trunk
- **Router Configuration:**
 - Create subinterfaces for each VLAN:
 - interface Fa0/0.20
 - encapsulation dot1Q 20
 - ip address 192.168.20.1 255.255.255.0
 - interface Fa0/0.30
 - encapsulation dot1Q 30

- ip address 192.168.30.1 255.255.255.0

5. Verification and Testing

- **Ping Test:**
 - Use the ping command to test connectivity between devices in different VLANs.
 - Example: ping 192.168.30.10 from a device in VLAN 20.
- **Routing Table:**
 - Check the router's routing table to verify routes:
 - show ip route

6. Key Points to Remember

- **VLAN Segmentation:** VLANs logically segment a network into different broadcast domains.
- **Inter-VLAN Routing:** Enables communication between VLANs using a router.
- **Traditional vs. Router-on-a-Stick:**
 - Traditional uses multiple interfaces.
 - Router-on-a-Stick uses a single interface with subinterfaces.
- **Configuration:**
 - Assign VLANs to switch ports.
 - Configure router interfaces or subinterfaces as default gateways.

7. Quick Tips for Viva

1. **Understand VLANs:** Know how VLANs segment networks and why inter-VLAN routing is needed.
2. **Practice Configurations:** Be familiar with configuring both traditional and router-on-a-stick methods.
3. **Explain the Process:** Be ready to explain how traffic flows between VLANs using a router.
4. **Verification:** Use ping and show ip route to verify configurations.